Durham College	
Course	**INFT1101: Wireless Network Security**
Professor	**Nargis Khan**
Lab2	**Configuring and Analyzing Wireless Router and Access Point**

This simple lab setup in Cisco Packet Tracer will help you to understand the basics of wireless network configuration and security.

Lab Objective

1. Understand basic wireless networking concepts.
2. Configure a wireless router.
3. Set up a secure wireless network using WPA2.
4. Set up a wired network
5. Explore network monitoring tools within Packet Tracer.
6. Extend communication range by using an AP.

Lab Setup

Equipment Needed

- **Cisco Packet Tracer** installed on your computer.
- **Devices**:
  - 1 Wireless Router (Cisco Router)
  - 2 PCs (to act as wired clients)
  - 2 Laptops (wireless client)
  - 1 Wireless Access Point (for extended coverage)
  - 1 Tablet , 1 laptop (as an additional wireless client connected with AP)

Lab Activities

Step 1: Create the Network Topology

1. **Open Cisco Packet Tracer**.
2. **Add Devices**:
   - Drag and drop a **Wireless Router (WRT300N-under wireless devices)** onto the workspace.
   - Add **two PCs** (PC1 and PC2) and 3 **Laptops(Laptop0,laptop1 and Laptop2)**.
   - Add a **Wireless Access Point**(AP-PT)for extended coverage
   - A **Tablet** as an additional wireless client.

3. **Connect Devices**: Follow figure 1
   - Connect PC1 and PC2 to the wireless router using **Ethernet cables**.(Under connections solid black line)
   - Connect Laptop1 and Laptop2 to the router wirelessly.
   - Connect AP to the router using Ethernet cable
   - Connect Laptop 2 and Table 1 to the AP wirelessly

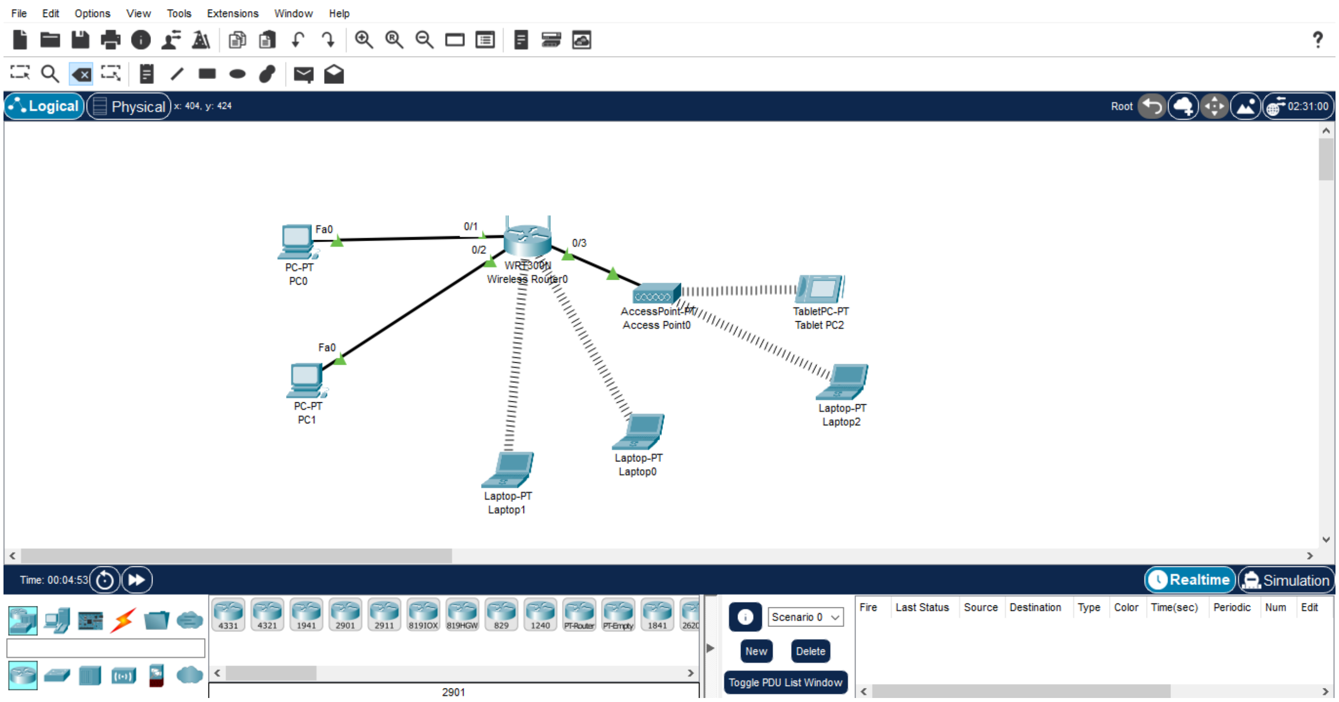


Figure 1

Step 2: Configure the Wireless Router

1. **Access the Router Configuration**:
   - Click on the wireless router and go to the **Config** tab.
2. **Configure the Router Settings**:
   - **Basic Settings (LAN)**:
     - Set the **Router Name** (e.g., WirelessRouter).
     - Assign an **IP Address** to the router (LAN e.g., 192.168.1.1) with a subnet mask of 255.255.255.0. This IP address will be use as a defaut gateway
3. **Configure the Wireless Settings**:
   - Navigate to the **Wireless** settings.
   - **SSID**: Set a unique SSID (e.g., MySecureNetwork).
   - **Security Mode**: Select WPA2-PSK.
     - Set a **Pre-Shared Key** (e.g., SecurePassword).

4. **Save Configuration**:
   - Ensure to save your settings before exiting.

Step 3: Configure the PCs and Laptop

1. **Configure PC1 and PC2**:
   - Click on each PC(PC1 and PC2) and go to the **Desktop** tab.
   - Click on **IP Configuration** and set the following:
     - **IP Address(static)**: 192.168.1.2 for PC1, 192.168.1.3 for PC2
     - **Subnet Mask**: 255.255.255.0
     - **Default Gateway**: 192.168.1.1
2. **Configure the Laptop**:
   - Click on each Laptop (Laptop0 and Laptop1) and go to the **Desktop** tab.
   - Click on **PC Wireless**:
     - Ensure the Wireless is turned on. To turn on wireless go to the **Physical tab**
       1. **Choose WPC300N**
       2. **Turn power off (yellow button)**
       3. **Remove the Ethernet card (black long slot PT Laptop-NM-1CEF)**
       4. Place wireless card WPC300N their.
       5. Turn the power on.
     - Select the SSID you configured (MySecureNetwork) under **Connect tab** and click connect
     - Enter the Pre-Shared Key (SecurePassword) when prompted.
     - Laptop1 and Laptop 2 will receive IP addresses dynamically from DHCP server.

Step 4: Configure the Access Point

1. **Access the Configuration**:
   - Click on the access point and go to the **Config** tab.
2. **Configure the Settings**:
   - **Basic Settings** :
     - Set the **Access Point Name** (e.g., Access Point 2).
3. **Configure the Port 1 Settings**:
   - Navigate to the **Port 1**.
   - **SSID**: Set a unique SSID (e.g., ExtendedAP).
   - **Security Mode**: Select WPA2-PSK.
     - Set a **Pre-Shared Key** (e.g., SecurePassword).

Step 5: Configure the Tablet:

1. **Access the Configuration**:
   - Click on the Tablet and go to the **Config** tab.

2. **Configure the Settings**:
    - **Basic Settings** : DHCP should be selected

3. **Configure the wireless0 Settings**:

    - **SSID**: Set SSID as ExtendedAP
    - **Security Mode**: Select WPA2-PSK.
        1. Set a **Pre-Shared Key** (e.g., SecurePassword).
    -

Step 6: Configure Laptop2 with Access point by following the same procedure as step 3.

Step 7: Testing Connectivity

1. **Test Wired Connections**:
    - From PC0, open the command prompt from the **Desktop** tab and ping PC1:

      ```
      ping 192.168.1.3
      ```

    - Ensure you receive replies.
    - From PC1, open the command prompt from the **Desktop** tab and ping PC0:

      ```
      ping 192.168.1.2
      ```

    - Ensure you receive replies.

Output 1: Take a screenshot of the successful ping results from each PC's pinging other PC **and attach here. [A Total of 2 screen shots]**

2. **Test Wireless Connection**:
    - From the Laptop0, open the command prompt and ping the router:

      ```
      ping 192.168.1.1
      ```

    - Do the same for Laptop1
    - Ping laptop1 from laptop0 and do the vice versa for laptop1
    - Ensure you receive replies.

Output 2: Take a screenshot of the successful ping results from each Laptop pinging other Laptop **and router and attach here. [A Total of 2 screen shots]**

3. **Test Connectivity on Extended wireless network**
    - Test the connection from Tablet by pinging the router and Laptop 2.
    - Test the connection from laptop2 by pinging the router and Tablet.
    - Ensure you receive replies.

Output 3: Take screenshots of the successful ping results as follows:

1. **From the Tablet:**
    - Ping the **Router** and **Laptop 2**.
2. **From Laptop 2:**
    - Ping the **Router** and the **Tablet**.

Please attach the screenshots here. **(A total of 2 screenshots, each showing the results of 2 pings.)**

Output 4: What is the role of the Access Point (AP) in our network setup, considering that it is connected to the wireless router and provides wireless connectivity to the Tablet and Laptop 2? Justify your answer.

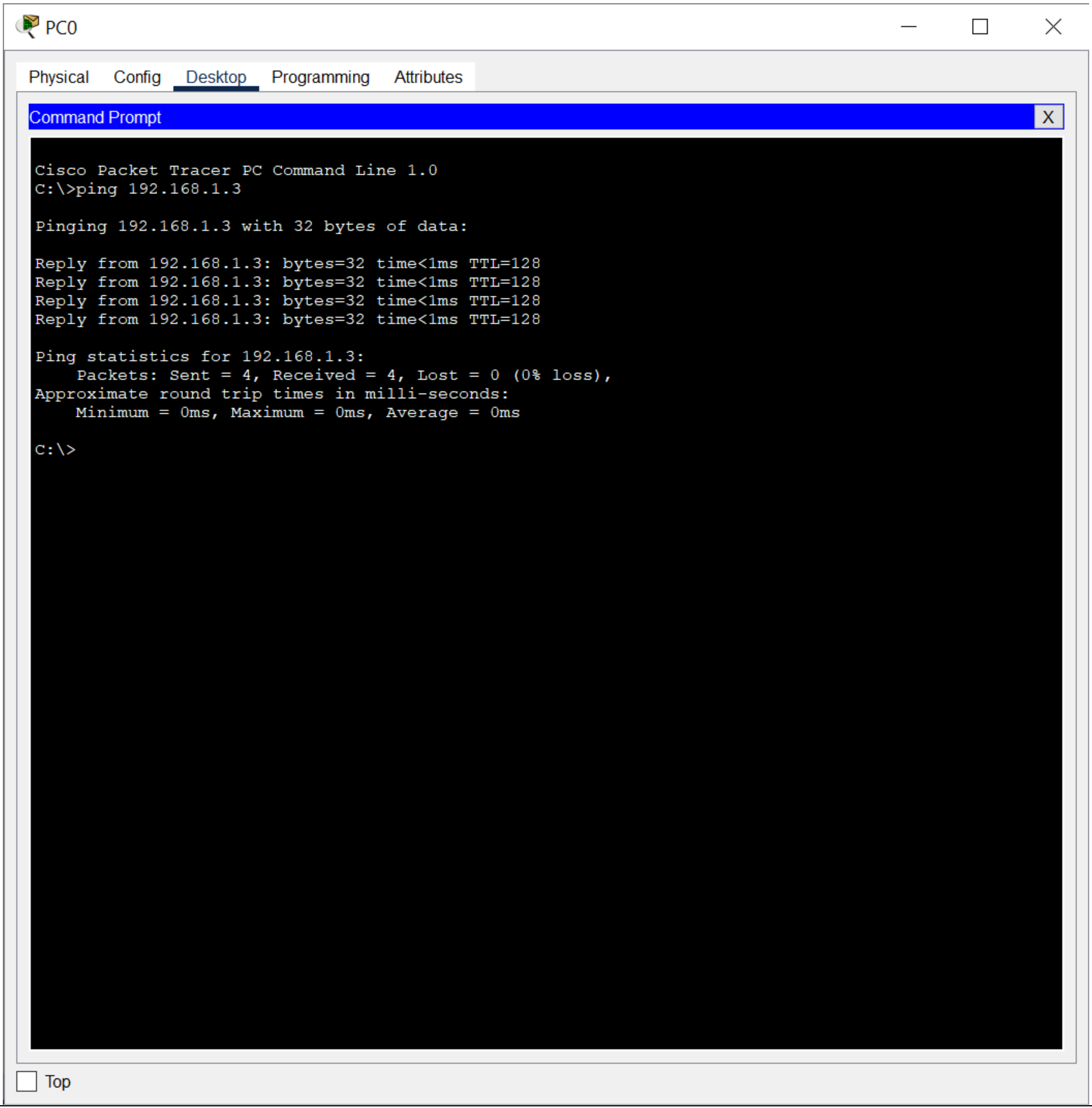
Output 5: Take the screen shot of the final Topology with your name and ID on top of the file.

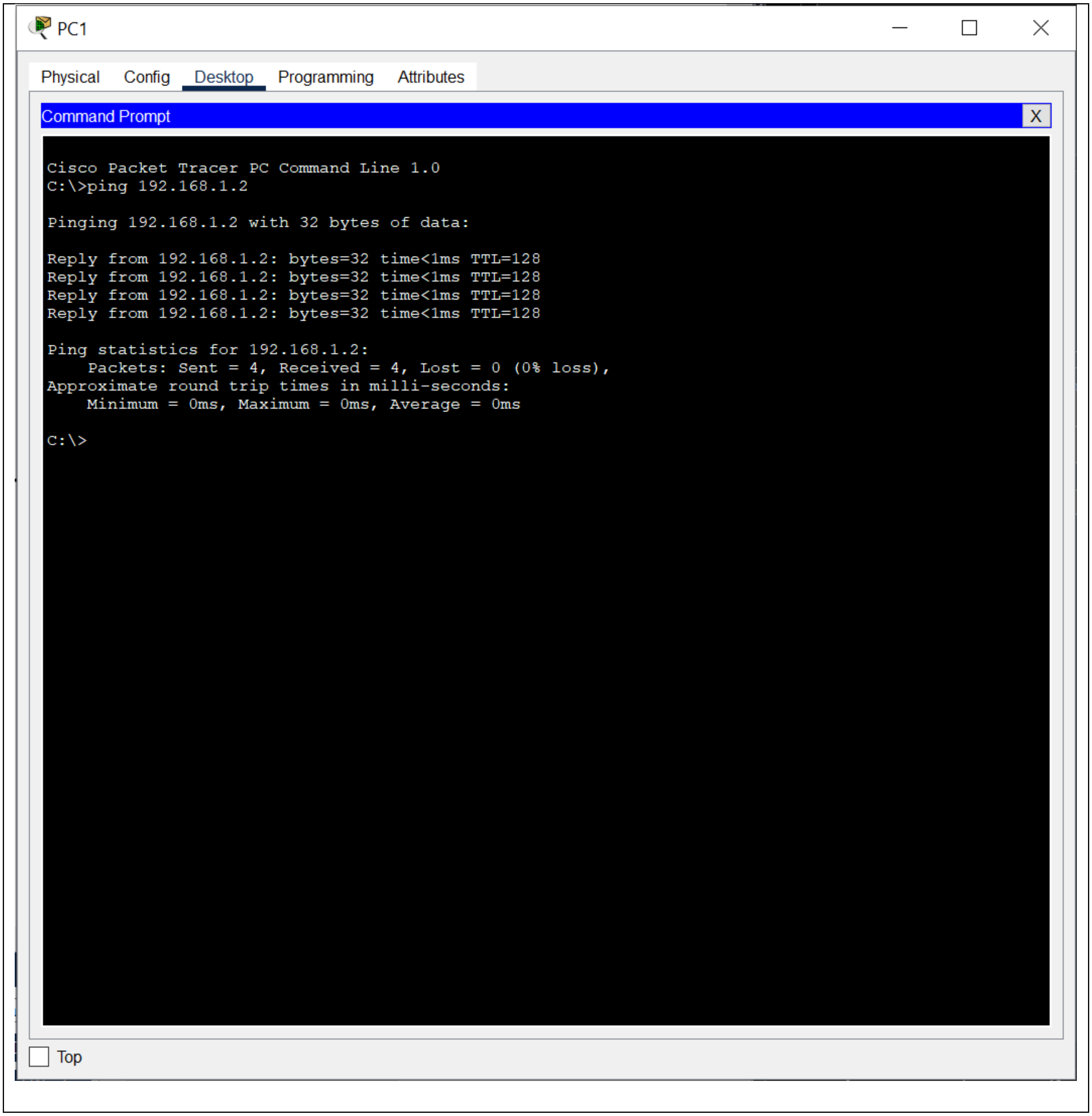
IT IS ONLY NECESSARY TO HAND IN THE FOLLOWING PAGES WITH SCREEN PRINTS .

Student Name	Melvin Viado
Performed Date	Thursday, September 25, 2025

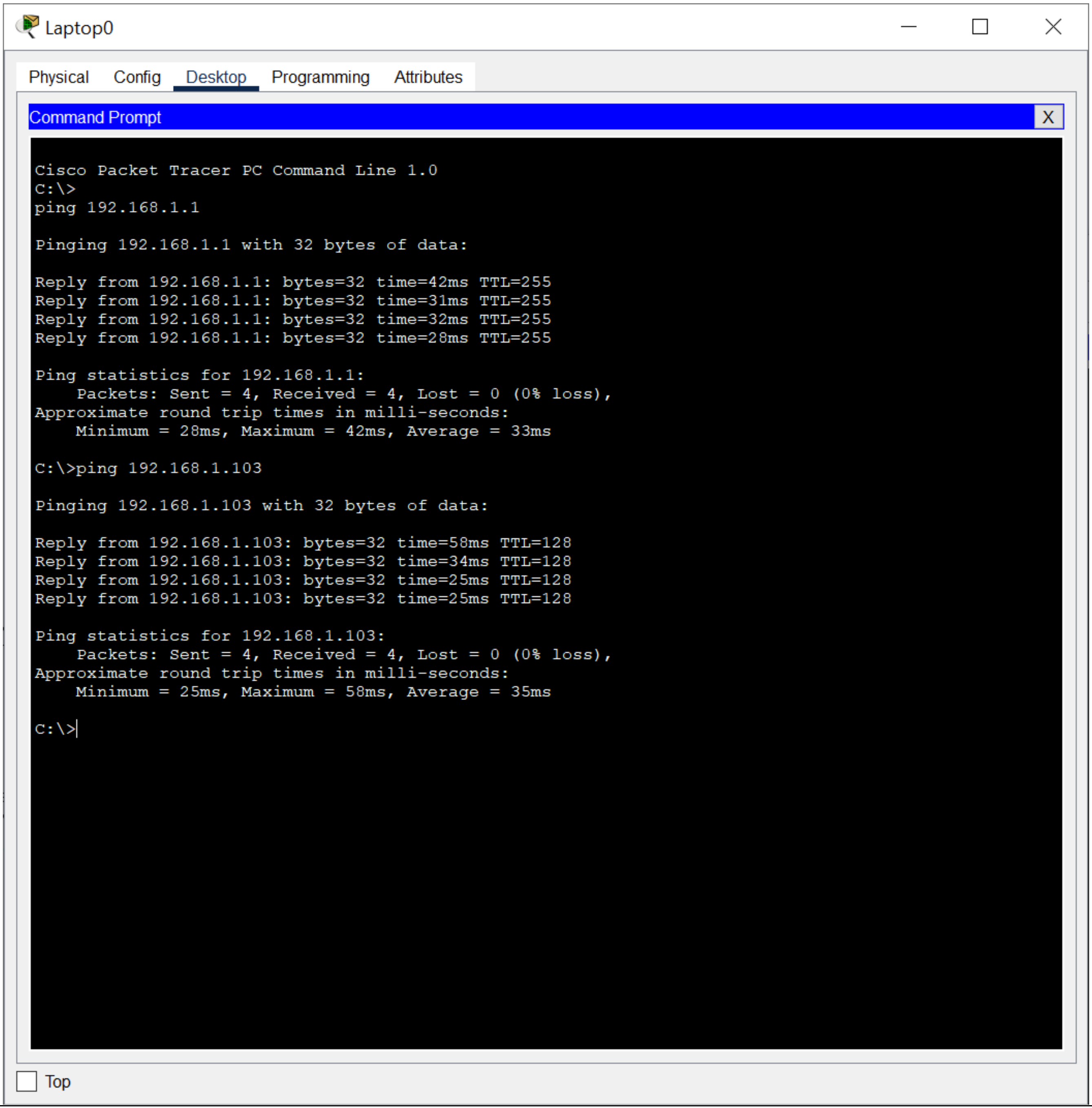
Output Section

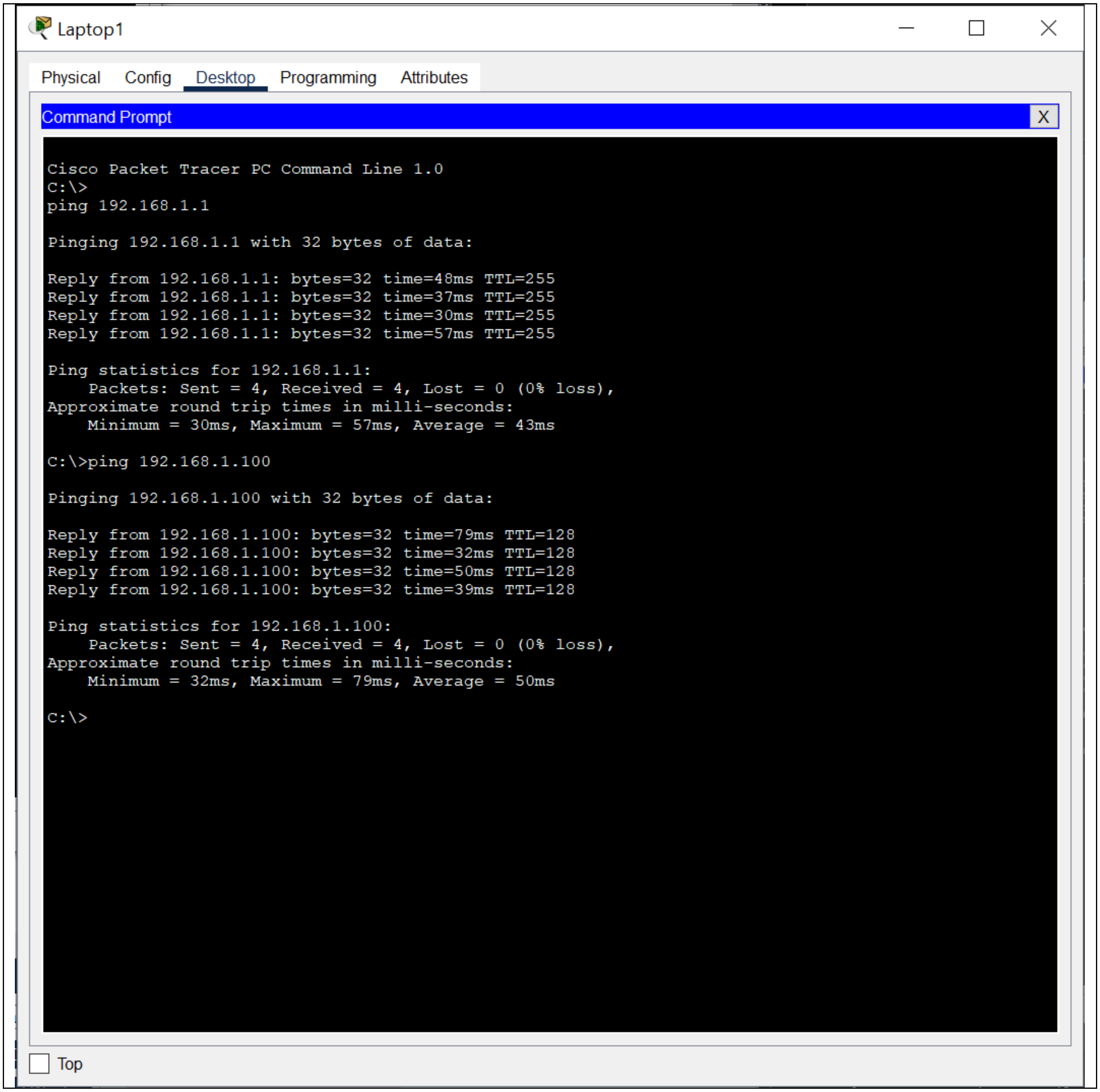
Output#1: PC1 and PC2 Successful Ping Results



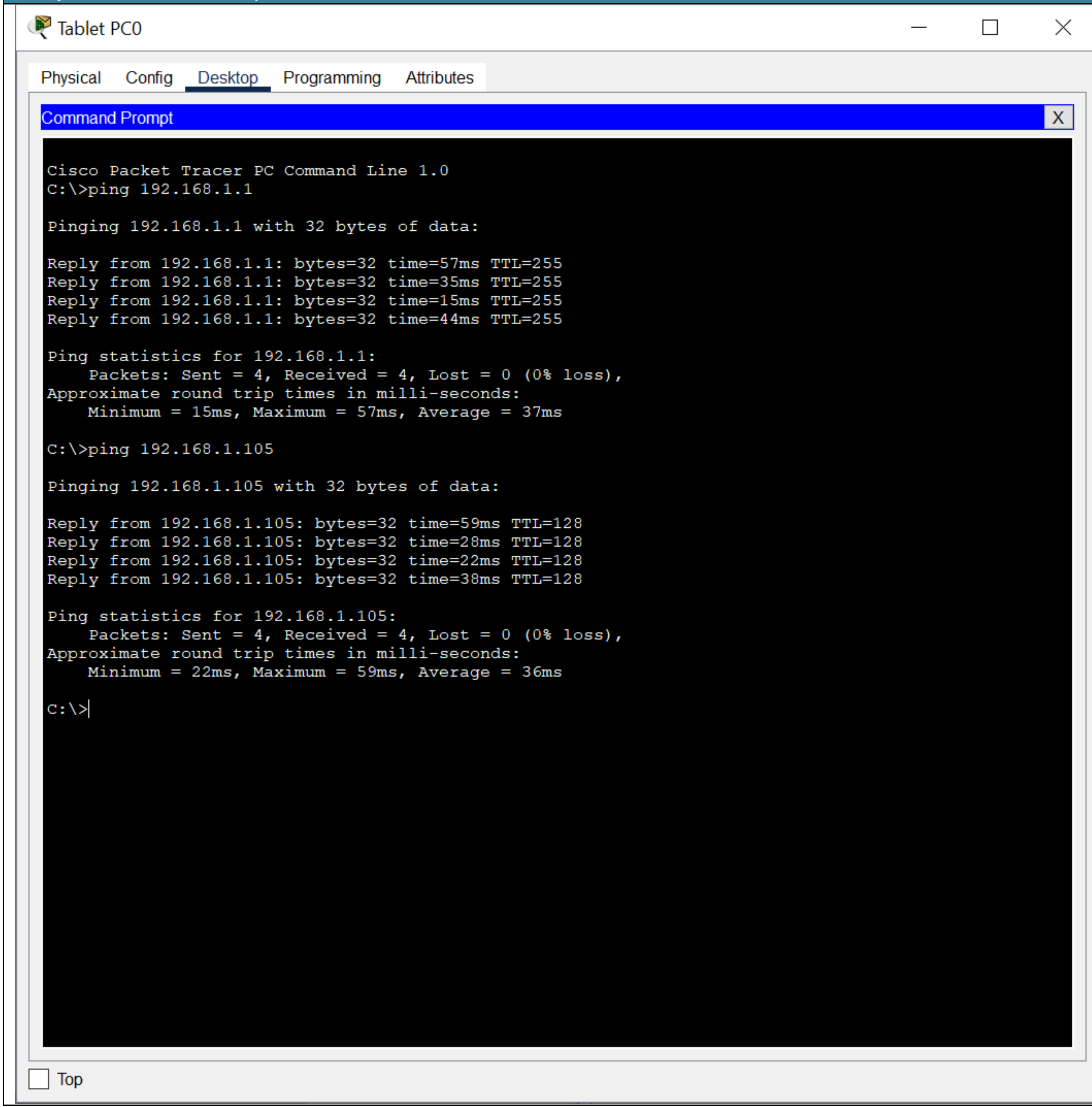

PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top

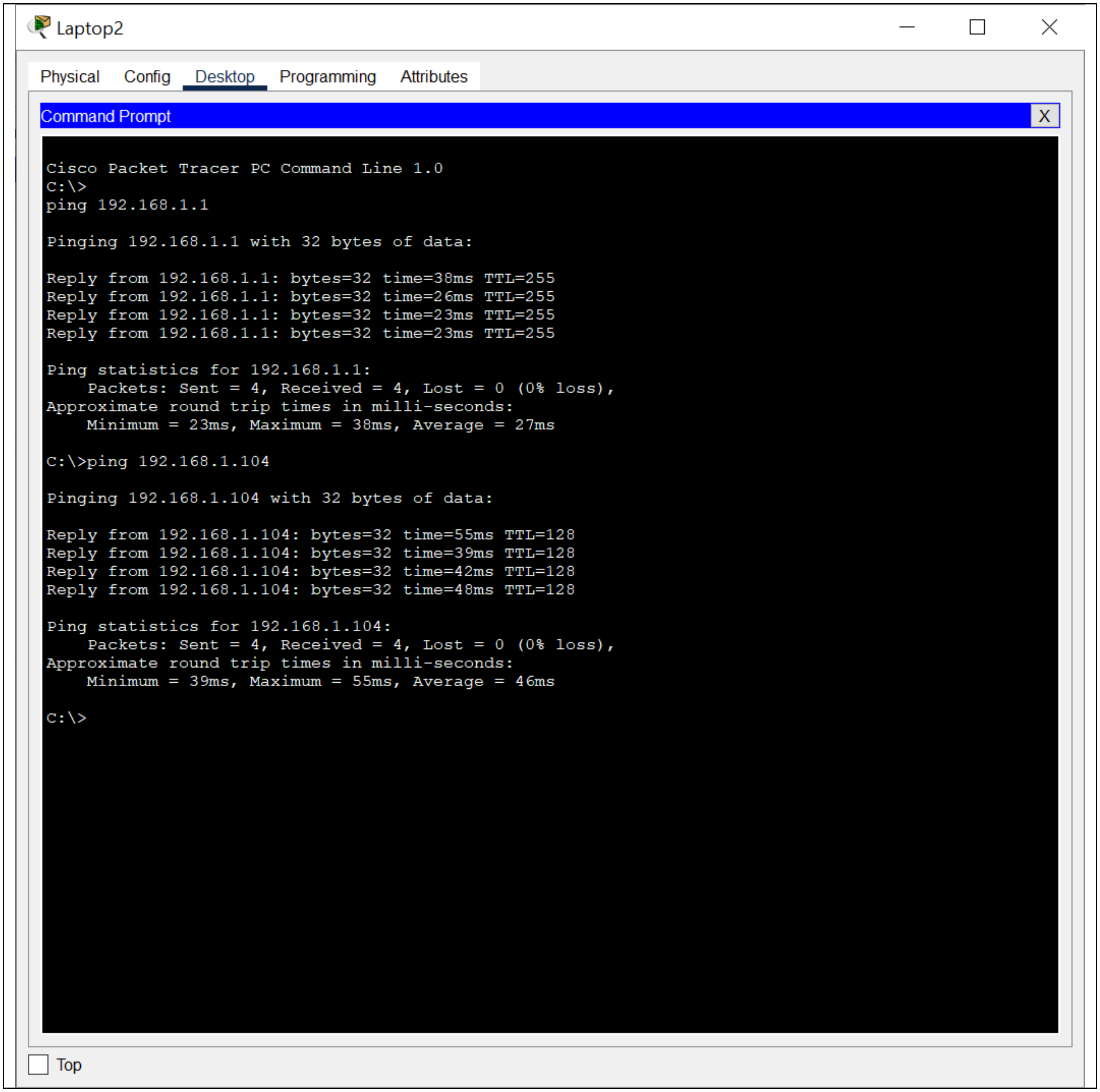
Output#2: Laptop Successful Ping Results from Other Laptop and Router




Laptop1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=48ms TTL=255
Reply from 192.168.1.1: bytes=32 time=37ms TTL=255
Reply from 192.168.1.1: bytes=32 time=30ms TTL=255
Reply from 192.168.1.1: bytes=32 time=57ms TTL=255
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 30ms, Maximum = 57ms, Average = 43ms
C:\>ping 192.168.1.100
Pinging 192.168.1.100 with 32 bytes of data:
Reply from 192.168.1.100: bytes=32 time=79ms TTL=128
Reply from 192.168.1.100: bytes=32 time=32ms TTL=128
Reply from 192.168.1.100: bytes=32 time=50ms TTL=128
Reply from 192.168.1.100: bytes=32 time=39ms TTL=128
Ping statistics for 192.168.1.100:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 32ms, Maximum = 79ms, Average = 50ms
C:\>
Top

Output#3: Connectivity on Extended Network Access Point




Laptop2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=38ms TTL=255
Reply from 192.168.1.1: bytes=32 time=26ms TTL=255
Reply from 192.168.1.1: bytes=32 time=23ms TTL=255
Reply from 192.168.1.1: bytes=32 time=23ms TTL=255

Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 23ms, Maximum = 38ms, Average = 27ms

C:\>ping 192.168.1.104

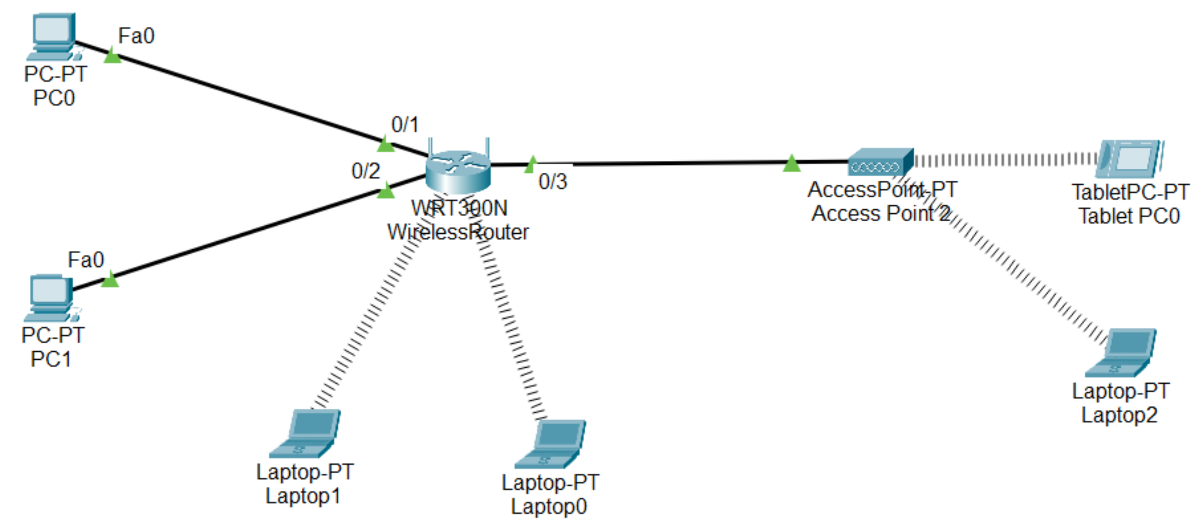
Pinging 192.168.1.104 with 32 bytes of data:

Reply from 192.168.1.104: bytes=32 time=55ms TTL=128
Reply from 192.168.1.104: bytes=32 time=39ms TTL=128
Reply from 192.168.1.104: bytes=32 time=42ms TTL=128
Reply from 192.168.1.104: bytes=32 time=48ms TTL=128

Ping statistics for 192.168.1.104:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 39ms, Maximum = 55ms, Average = 46ms

C:\>
Top

Output#4: {Put appropriate caption/title here}
The role of the access point in our network setup is to extend the wireless network coverage. The tablet and laptop may be physically located closer to the access point and further from the wireless router. The signal of the router will be weak for these devices while the signal of the access point will be strong.

Output#5: Melvin Viado - 100899671


Marking Scheme

Output 1		3	2	1	0
Output 2		3	2	1	0
Output 3		3	2	1	0
Output 4	4	3	2	1	0
Output 5		3	2	1	0
This Page included				1	0
Output Titles				1	0
Total:					/18